

Empirical Investigation of Multi-Sensor Fusion, Representation Collapse, and Policy Stability in CARLA Reinforcement Learning

Bachelor Thesis Project (BTP)
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Project Codebase: <https://github.com/MSR327/vlm>

Abstract—Deep reinforcement learning (RL) for vision-based autonomous driving requires balancing perceptual coverage, state dimensionality, and policy optimization stability. We present a systematic empirical study on sensor scaling and representation learning in the CARLA simulator using Proximal Policy Optimization (PPO). Rather than reporting only an idealized endpoint, we evaluate an empirical ladder across four sensory configurations: (1) a monocular baseline (100-dim) achieving 39.0% route completion (195 m, $R = 313.21 \pm 38.45$, lane deviation 0.41 ± 0.16 m) in Town01 but failing at sharp intersections (Meter 190) and collapsing under zero-shot transfer in Town02 (1.0% completion, 100% collisions); (2) a multi-view forward surround rig with 3 cameras (290-dim, $0^\circ, \pm 60^\circ$) recovering completion to 25.0% (125 m, $R = 158.31 \pm 26.80$, lane deviation 0.58 ± 0.24 m) by providing a 210° panoramic forward arc; (3) a naive 4-camera panoramic surround rig (385-dim) that induces policy standstill collapse (3.4% completion, 17 m, $R = 17.35 \pm 3.12$, lane deviation 0.08 ± 0.03 m) due to counter-directional optic flow, 385-dim gradient dissipation, and DirectX 11 buffer overruns; and (4) an orthogonal multimodal fusion rig (195-dim) pairing front vision with a 2D Bird’s-Eye-View (BEV) LiDAR height grid, achieving monotonic reward progression ($-8.81 \rightarrow -5.02 \pm 4.18$, peak $+19.84$, lane deviation 0.82 ± 0.43 m) with zero stalls. Furthermore, we isolate and resolve two severe numerical instabilities in sparse BEV autoencoding: a float32 exponential overflow (NaN loss) cured via bounded log-sigma clipping $[-15.0, 1.0]$, and a 1.8×10^6 latent variance collapse cured via unconditional forward execution. Finally, we analyze the structural limits of static latent concatenation across independent VAEs, proving that ungrounded MLP policy heads lack spatial cross-attention mechanisms necessary to reconcile multi-modal features, establishing a principled empirical motivation for patch-based Vision-Language-Action (VLA) and cross-attention architectures.

Index Terms—Autonomous driving, CARLA simulator, sensor fusion, Bird’s-Eye-View LiDAR, Proximal Policy Optimization, Variational Autoencoder, representation learning, cross-attention.

I. INTRODUCTION & MOTIVATION FOR MULTI-SENSOR SCALING

End-to-end autonomous driving aims to map onboard perceptual sensor streams directly to continuous steering, throttle, and brake commands [1]–[3]. While foundational imitation learning and reinforcement learning agents operated primarily on a single forward-facing RGB camera, navigating real-world or simulated urban environments demands wider spatial perception and reliable 3D distance awareness [4], [6].

A. Engineering Motivation for Progressive Sensor Scaling

Deploying autonomous agents in urban environments exposes four fundamental physical and geometric constraints of single-sensor architectures:

- 1) **Geometric Blind Spots at Urban Intersections:** Standard monocular perspective cameras provide a horizontal field of view (FOV) of 90° – 125° . While sufficient for straight highway tracking, this geometry creates complete perceptual blindness to lateral corridors ($> 60^\circ$ off-center). At sharp 90° urban turns or T-junctions, the inner lane boundary and road curvature exit the front camera frame before the vehicle can physically rotate its yaw axis, leaving the control policy without guidance.
- 2) **Ill-Posed Inverse Depth Ambiguity:** Projecting 3D physical coordinates (x, y, z) onto a 2D pixel array (u, v) discards metric range. Monocular depth estimation is fundamentally ill-posed and susceptible to scale ambiguity, dynamic illumination shifts, rain spray, and sun glare [16].
- 3) **Occlusion Resilience in Dense Traffic:** In dense traffic, leading trucks, buses, or street furniture occlude the forward path. Multi-view and surround camera arrays provide parallax and line-of-sight redundancy around obstacles.
- 4) **Photometric vs. Geometric Complementarity:** While vision provides dense semantic classification (lane markings, traffic signs, stop lines), active LiDAR sensors provide metric, illumination-invariant spatial ranges with centimeter precision. Fusing camera vision with LiDAR Bird’s-Eye-View (BEV) representations combines semantic context with absolute metric geometry.

However, simply adding more sensors naively into reinforcement learning pipelines frequently degrades performance rather than improving it. In this work, we construct and evaluate an empirical four-stage *sensor ladder* in CARLA to investigate the trade-offs between perceptual coverage, state dimensionality, simulation stability, and policy convergence.

II. RELATED WORK

A. Sensor Fusion in Autonomous Driving

Learning by Cheating (LBC) [6] proved that monocular camera agents suffer from severe spatial myopia, whereas agents trained on privileged Bird’s-Eye-View (BEV) abstractions achieve substantially higher route completion. *TransFuser* [5] demonstrated that combining multi-view cameras with LiDAR point clouds via cross-attention outperforms single-modality baselines by fusing optical semantic segmentation with invariant metric 3D geometry. *InterFuser* [7] extended multi-sensor fusion across panoramic perspectives to eliminate blind spots and ensure collision safety in complex traffic scenarios.

B. Bird’s-Eye-View (BEV) Point Cloud Representations

Projecting raw 3D LiDAR point clouds into 2D Bird’s-Eye-View (BEV) spatial grids preserves metric physical distances while enabling standard 2D convolutional backbones to extract geometric affordances efficiently: *PointPillars* [8] pioneered vertical pillar tokenization for real-time 3D object detection from point clouds. *BEVDet* [9] further established that transforming perceptual features into an ego-centric 2D metric ground plane provides superior spatial localization and trajectory prediction compared to perspective-view processing.

C. Representation Learning & Dimensionality in RL

Model-free reinforcement learning directly from raw high-dimensional pixels suffers from extreme sample inefficiency and optimization instability [10], [12]. *CARL* [10] demonstrated in CARLA benchmark evaluations that policy gradient convergence is heavily constrained by observation representation dimensionality. *Roach* [11] addressed high-dimensional sample inefficiency by pre-training policies over privileged BEV affordance masks. Variational Autoencoders (VAEs) [15] are widely deployed to compress high-dimensional observation frames into compact, regularized continuous latent vectors $\mathbf{z} \in \mathbb{R}^{d_z}$ to feed downstream policy heads. Furthermore, abstracting natural RGB streams into Cityscapes semantic class palettes [16] removes extraneous variance originating from dynamic weather conditions, sun glare, and high-frequency textures.

D. Policy Optimization & Control Stability

Proximal Policy Optimization (PPO) [12] represents the primary on-policy clipped surrogate reinforcement learning objective for continuous control. Generalized Advantage Estimation (GAE- λ) [13] introduces exponential discounting to calibrate the bias-variance trade-off in policy advantage estimates. Conditioning for Action Policy Smoothness (CAPS) [14] demonstrated that model-free continuous RL agents frequently suffer from high-frequency actuator chatter and aggressive steering jitter unless explicitly regularized via temporal action smoothing and smoothness penalties.

III. METHODOLOGY

A. Reinforcement Learning Formulation

We formulate the closed-loop autonomous driving problem as an episodic Partially Observable Markov Decision Process (POMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **Control Frequency:** The agent interacts synchronously with the CARLA simulator at a fixed frequency of $f = 20$ Hz ($\Delta t = 0.05$ s).
- **Action Space:** The policy outputs a continuous 2-dimensional action vector $\mathbf{a}_t = [a_{\text{steer}}, a_{\text{long}}] \in [-1, 1]^2$.
- **Decoupled Longitudinal Control:** To prevent simultaneous, conflicting throttle and brake commands, we implement a signed longitudinal mapping:

$$\text{throttle}_t = \max(a_{\text{long}}, 0.0) \quad (1)$$

$$\text{brake}_t = \max(-a_{\text{long}}, 0.0) \quad (2)$$

- **Temporal Action Smoothing:** Guided by the CAPS framework [14], we filter steering predictions using an Exponential Moving Average (EMA) to prevent vehicle instability and actuator chatter:

$$\delta_t = 0.6 \delta_{t-1} + 0.4 a_{\text{steer},t} \quad (3)$$

where δ_t denotes the physical steering angle applied to the CARLA ego-vehicle.

- **Multi-Objective Reward Function:** At each discrete time step t , the agent receives a shaped reward:

$$r_t = R_{\text{speed}} \cdot R_{\text{center}} \cdot R_{\text{angle}} + R_{\text{penalty}} \quad (4)$$

where the individual components are defined as:

$$R_{\text{speed}} = \min(v_t/22.0, 1.0) \quad (5)$$

$$R_{\text{center}} = \max\left(1.0 - \frac{d_{\text{center}}}{3.0}, 0.0\right) \quad (6)$$

$$R_{\text{angle}} = \max\left(1.0 - \frac{|\theta_{\text{heading}}|}{20^\circ}, 0.0\right) \quad (7)$$

Here v_t represents scalar forward speed in km/h, d_{center} denotes signed perpendicular lane-center deviation in meters, and θ_{heading} represents the angular heading error relative to the lane tangent. Dynamic collisions, sidewalk encroachments, or lane deviations exceeding 3.0 m terminate the episode with a penalty of $R_{\text{penalty}} = -10.0$.

B. The Empirical Sensor Ladder

To isolate the empirical trade-offs between perceptual coverage, representation dimensionality, and policy stability, we evaluate four systematic sensor rungs (Fig. 1):

- 1) **Rung 1: Monocular Baseline (100-dim):** A single front-facing semantic camera (0° yaw, 125° FOV). The state vector concatenates the 95-dimensional visual latent with a 5-dimensional navigation telemetry vector:

$$\mathbf{s}_t = [\mathbf{z}_{\text{cam}} \in \mathbb{R}^{95} \parallel \mathbf{x}_{\text{nav}} \in \mathbb{R}^5] \in \mathbb{R}^{100} \quad (8)$$

where $\mathbf{x}_{\text{nav}} = [v_t, d_{\text{center}}, \theta_{\text{heading}}, \delta_{t-1}, a_{\text{long},t-1}]$.

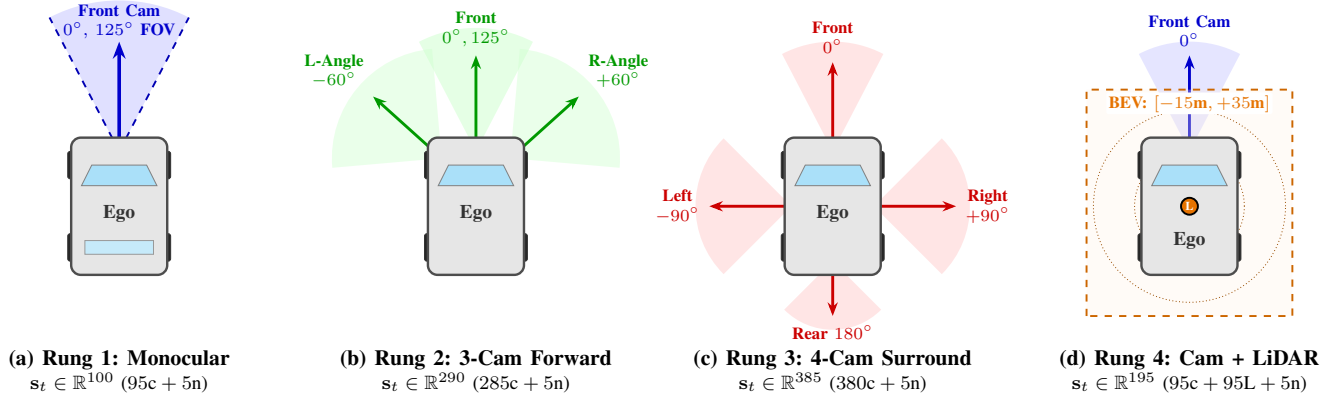


Fig. 1. Architectural diagrams of the four sensor ladder configurations evaluated in CARLA. (a) Rung 1: Monocular front baseline. (b) Rung 2: Multi-view forward surround rig with 3 cameras (210° panoramic forward arc). (c) Rung 3: Naive 4-camera panoramic surround rig (360° horizontal coverage). (d) Rung 4: Orthogonal multimodal fusion of front camera and 2D BEV LiDAR.

- 2) **Rung 2: Multi-View Forward Surround (290-dim):** To address the severe turn blindness of the monocular baseline while preserving forward motion dynamics, the rig integrates three forward-focused cameras: Front (0° yaw, 125° FOV), Left-Angled (-60° yaw, 90° FOV), and Right-Angled (+60° yaw, 90° FOV), yielding a combined forward panoramic arc of 210°:

$$s_t = [z_{\text{front}} \parallel z_{\text{left}} \parallel z_{\text{right}} \parallel x_{\text{nav}}] \in \mathbb{R}^{290} \quad (9)$$

- 3) **Rung 3: Naive 4-Camera Panoramic Surround (385-dim):** Four orthogonal semantic cameras covering the full 360° horizontal perimeter: Front (0°, 125° FOV), Left (-90°, 90° FOV), Right (+90°, 90° FOV), and Rear (180°, 90° FOV). The state is formed by direct latent concatenation:

$$s_t = [z_{\text{front}} \parallel z_{\text{left}} \parallel z_{\text{right}} \parallel z_{\text{rear}} \parallel x_{\text{nav}}] \in \mathbb{R}^{385} \quad (10)$$

- 4) **Rung 4: Orthogonal Multimodal Fusion (195-dim):** One front-facing camera (0°, 125° FOV) combined with a roof-mounted 3D LiDAR projected into a 2D Bird’s-Eye-View (BEV) height/occupancy grid:

$$s_t = [z_{\text{cam}} \in \mathbb{R}^{95} \parallel z_{\text{lidar}} \in \mathbb{R}^{95} \parallel x_{\text{nav}} \in \mathbb{R}^5] \in \mathbb{R}^{195} \quad (11)$$

C. Simulation Environment, Vehicle Physics, & Benchmark Maps

To guarantee experimental repeatability and eliminate physics artifacts, all experiments adhere to a rigorously controlled simulation environment:

- **Simulation Platform:** CARLA version 0.9.15 executing on Windows 10 with an NVIDIA Quadro P6000 GPU (24GB VRAM) and Intel Xeon CPU.
- **Synchronous Execution & Physics Substepping:** Fixed simulation time step $\Delta t = 0.05$ s (20 Hz). Server and client operate in non-real-time synchronous mode to ensure zero frame dropping. Under the PhysX engine,

substepping is configured at 100 Hz ($\delta t_{\text{sub}} = 0.01$ s, 5 substeps per step) to guarantee rigid-body stability and consistent tire contact forces.

- **Ego-Vehicle Dynamics:** Standard four-wheeled passenger vehicle asset (Tesla Model 3 / Audi TT) with curb mass $m = 1500$ kg, wheelbase $L = 2.87$ m, and dry asphalt surface tire friction coefficient $\mu = 1.0$.
- **Sensor Mounting Coordinates (Extrinsics):**
 - *Front Camera:* Mounted on the windshield header at $(x = 2.0 \text{ m}, y = 0.0 \text{ m}, z = 1.4 \text{ m})$, orientation (pitch = -5°, yaw = 0°, roll = 0°), 125° FOV.
 - *Left-Angled Camera:* $(x = 2.0 \text{ m}, y = -0.2 \text{ m}, z = 1.4 \text{ m})$, orientation (pitch = -5°, yaw = -60°, roll = 0°), 90° FOV.
 - *Right-Angled Camera:* $(x = 2.0 \text{ m}, y = +0.2 \text{ m}, z = 1.4 \text{ m})$, orientation (pitch = -5°, yaw = +60°, roll = 0°), 90° FOV.
 - *Rear Camera (Rung 3):* $(x = -1.5 \text{ m}, y = 0.0 \text{ m}, z = 1.4 \text{ m})$, orientation (pitch = -5°, yaw = 180°, roll = 0°), 90° FOV.
 - *3D LiDAR Sensor:* Mounted on the roof center at $(x = 0.0 \text{ m}, y = 0.0 \text{ m}, z = 2.4 \text{ m})$, configured with 32 vertical laser channels spanning -30° to +10°, rotation frequency 20 Hz, outputting 100,000 points/sec over a 50 m radius.
- **Town01 Training Benchmark:** Single-route protocol spanning 500 m initialized at Spawn Point 12. Road width is 6.0 m (3.0 m per lane), bordered by 0.15 m concrete curbs. The trajectory features a long 190 m straightaway leading directly into an unbanked, blind 90° right turn with turning radius $r \approx 12.5$ m (curvature $\kappa \approx 0.08 \text{ m}^{-1}$) at *Meter 190*.
- **Town02 Zero-Shot Transfer Benchmark:** Evaluated across 20 unseen evaluation episodes. Town02 features narrow 3.0 m lanes, high-density multi-story stone masonry facades, sharp 90° blind T-junctions, and complex building geometries ($> 1.2 \times 10^6$ polygons per frame).
- **Dynamic Traffic & Environmental Variations:** A

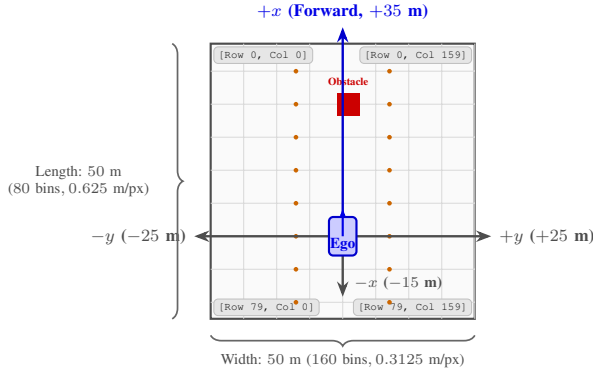


Fig. 2. Geometric coordinate mapping and binning discretization of the 2D Bird's-Eye-View (BEV) LiDAR projection raster (80×160 channels) centered egocentrically around the vehicle.

dataset of 10,000 multi-sensor frames was collected under 29 autonomous NPC vehicles operating via CARLA Traffic Manager (15–35 km/h) and 4 randomized weather presets (ClearNoon, HardRainNoon, WetSunset, CloudyNight).

D. 2D Bird's-Eye-View (BEV) LiDAR Projection Geometry

A 360-degree raycast LiDAR sensor is top-mounted on the ego-vehicle chassis at $(x = 0.0, y = 0.0, z = 2.4 \text{ m})$. At each time step, raw 3D point cloud returns $\mathcal{P} = \{(x_i, y_i, z_i)\}_{i=1}^N$ are projected into an asymmetric Bird's-Eye-View spatial grid (Fig. 2):

- 1) **Asymmetric Spatial Bounding:** We define an asymmetric region of interest biased longitudinally forward:

$$\mathcal{P}_{\text{filt}} = \{(x_i, y_i, z_i) \in \mathcal{P} \mid -15.0 \leq x_i \leq +35.0, -25.0 \leq y_i \leq +25.0\} \quad (12)$$

- 2) **Discretization to Pixel Grid Coordinates (u_i, v_i) :** The continuous spatial coordinates are mapped into an 80×160 discrete matrix:

$$u_i = \text{clip} \left(\left\lfloor \frac{35.0 - x_i}{50.0} \times 79 \right\rfloor, 0, 79 \right) \quad (13)$$

$$v_i = \text{clip} \left(\left\lfloor \frac{y_i + 25.0}{50.0} \times 159 \right\rfloor, 0, 159 \right) \quad (14)$$

- 3) **Normalized Height Assignment:** The elevation $z_i \in [-2.0 \text{ m}, +2.0 \text{ m}]$ is mapped to continuous unit range:

$$\tilde{z}_i = \text{clip} \left(\frac{z_i + 2.0}{4.0}, 0.0, 1.0 \right) \quad (15)$$

- 4) **Max-Pooling Channel Assignment:** Where multiple LiDAR reflections project into the same spatial pixel (u, v) , max-pooling preserves the highest obstacle return:

$$\mathbf{L}_{\text{bev}}(u, v) = \max_{i: u_i=u, v_i=v} \tilde{z}_i \quad (16)$$

yielding a single-channel raster $\mathbf{L}_{\text{bev}} \in [0, 1]^{80 \times 160}$.

TABLE I
LAYER SPECIFICATIONS OF THE VARIATIONAL AUTOENCODER

Sub-module	Layer Operator	Filter / Units	Output Shape
Encoder	Input Tensor	—	$160 \times 80 \times 3$
	Conv2D (stride 2, ReLU)	$32 \times (4 \times 4)$	$80 \times 40 \times 32$
	Conv2D (stride 2, ReLU)	$64 \times (3 \times 3)$	$40 \times 20 \times 64$
	Batch Normalization	—	$40 \times 20 \times 64$
	Conv2D (stride 2, ReLU)	$128 \times (4 \times 4)$	$20 \times 10 \times 128$
	Conv2D (stride 2, ReLU)	$256 \times (3 \times 3)$	$10 \times 5 \times 256$
Bottleneck	Flatten & Dense	1024 units	1024
	Linear Head (μ)	95 units	95
Decoder	Linear Head ($\log \sigma^2$)	95 units (clipped)	95
	Dense Projection	$1024 \rightarrow 12,800$ units	$10 \times 5 \times 256$
	Conv2DTranspose (s=2)	$128 \times (4 \times 4)$	$20 \times 10 \times 128$
	Conv2DTranspose (s=2)	$64 \times (3 \times 3)$	$40 \times 20 \times 64$
	Conv2DTranspose (s=2)	$32 \times (4 \times 4)$	$80 \times 40 \times 32$
	Conv2DTranspose (s=1)	$3 \times (3 \times 3)$, Sigm.	$160 \times 80 \times 3$

E. End-to-End System Pipeline & Architecture

Fig. 3 details the end-to-end signal flow from raw perception to low-level vehicle actuation. Camera and LiDAR data streams are compressed by frozen convolutional encoders, concatenated with vehicle kinematics, and processed by PPO actor-critic networks to issue smoothed continuous controls at 20 Hz.

F. Proximal Policy Optimization with GAE- λ

The policy network $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$ and state-value critic $V_\phi(\mathbf{s}_t)$ are optimized using PPO with Generalized Advantage Estimation:

- 1) **Temporal Difference Residual:**

$$\delta_t^V = r_t + \gamma V_\phi(\mathbf{s}_{t+1})(1 - d_t) - V_\phi(\mathbf{s}_t) \quad (17)$$

where $d_t \in \{0, 1\}$ indicates episodic termination.

- 2) **Generalized Advantage Estimation (GAE- λ):**

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V \quad (18)$$

with discount $\gamma = 0.99$ and smoothing parameter $\lambda = 0.95$.

- 3) **PPO Clipped Surrogate Objective:**

$$\mathcal{L}_{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (19)$$

where probability ratio $\rho_t = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t)}$ and clipping threshold $\epsilon = 0.2$.

- 4) **Value Function Mean Squared Error Loss:**

$$\mathcal{L}_{\text{VF}}(\phi) = \hat{\mathbb{E}}_t \left[(V_\phi(\mathbf{s}_t) - \hat{R}_t)^2 \right], \quad \hat{R}_t = \frac{R_t - \mu_R}{\sigma_R + 10^{-7}} \quad (20)$$

Standardizing returns $\hat{R}_t$ guarantees numerical stability across diverse episodic durations.

IV. IMPLEMENTATION DETAILS

A. Deep Neural Network Architectures

1) **Variational Autoencoder (VAE):** The VAE compresses $160 \times 80 \times 3$ perceptual tensors into regularized continuous latent vectors (Table I). To guarantee scientific fairness across the ladder, the LiDAR BEV autoencoder in Rung 4 adopts

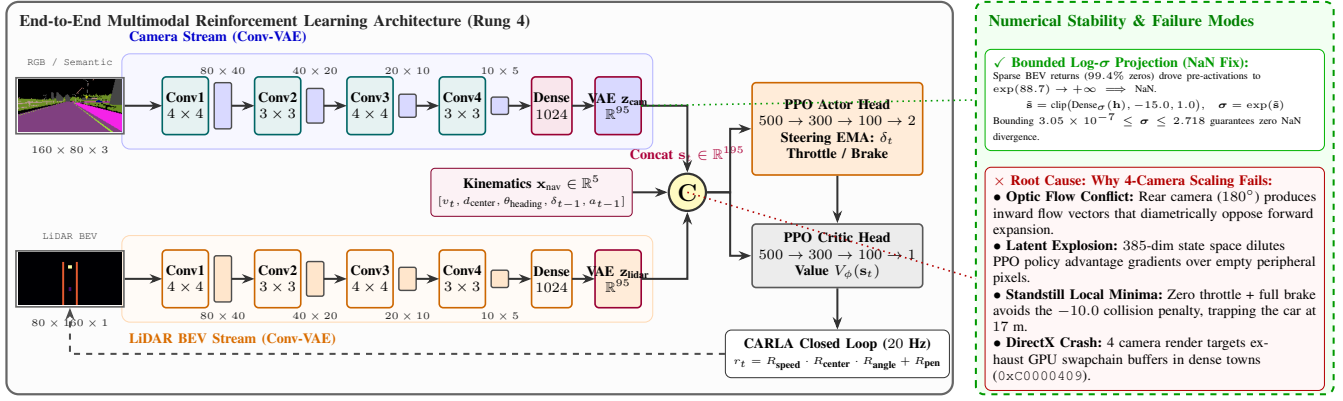


Fig. 3. End-to-end multimodal deep reinforcement learning architecture in CARLA, inspired by TransFuser [5]. Perspective RGB streams and 2D metric Bird’s-Eye-View (BEV) LiDAR grids are tokenized by frozen Conv-VAE encoders into compact latent vectors ($\mathbf{z} \in \mathbb{R}^{95}$), concatenated with kinematic vehicle telemetry, and fed to PPO actor-critic heads for continuous 20 Hz closed-loop control. The callout (right) details our bounded log-sigma stabilization preventing NaN explosions alongside the empirical failure mechanisms of naive 4-camera scaling.

the exact same layer geometry, parameter capacity (13.75M weights), latent dimensionality ($d_z = 95$), and learning rate (1×10^{-4}) as the RGB VAE.

2) *PPO Actor-Critic Policy*: Both actor and critic modules operate directly upon the combined sensory-navigational state $\mathbf{s}_t$:

- *Actor Network*: Multi-layer perceptron mapping $\mathbf{s}_t \rightarrow [500, \tanh] \rightarrow [300, \tanh] \rightarrow [100, \tanh] \rightarrow [2, \tanh]$.
- *Critic Network*: Multi-layer perceptron mapping $\mathbf{s}_t \rightarrow [500, \tanh] \rightarrow [300, \tanh] \rightarrow [100, \tanh] \rightarrow [1, \text{Linear}]$.
- *Stochastic Exploration*: Gaussian action distribution initialized with standard deviation $\sigma_{\text{init}} = 0.20$.

B. Numerical Stabilization on Sparse Point Clouds

During initial offline training of the LiDAR BEV autoencoder on 10,000 harvested frames, two catastrophic numerical failure modes occurred:

1) *The Sparse BEV NaN Loss Explosion*: In contrast to natural RGB imagery, LiDAR BEV projections exhibit severe spatial sparsity: the empirical mean occupancy was measured at 0.0059 (99.41% of pixels are exact zeros). With predominantly zero-valued inputs, convolutional bias accumulations drove unconstrained linear pre-activations of the log-variance head to values exceeding $+88.7$. In standard IEEE 754 single-precision floating point:

$$\exp(88.7228) \rightarrow +\infty \quad (21)$$

Multiplying $+\infty$ by stochastic Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ produced NaN values, causing gradient backpropagation to immediately corrupt all network weights. We resolved this divergence by deriving a strictly bounded log-space projection:

$$\tilde{\mathbf{s}} = \text{clip}(\text{Dense}_\sigma(\mathbf{h}), -15.0, 1.0), \quad \sigma = \exp(\tilde{\mathbf{s}}) \quad (22)$$

This mathematically guarantees:

$$3.05 \times 10^{-7} \leq \sigma \leq 2.718 \quad (23)$$

Coupled with gradient norm clipping ($\text{clipnorm} = 1.0$) and dynamic GPU memory allocation, the VAE training loss

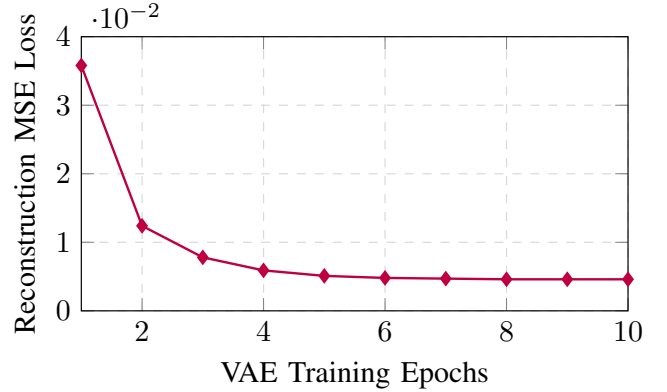


Fig. 4. Reconstruction mean squared error (MSE) loss during offline training of the 2D LiDAR BEV Variational Autoencoder, demonstrating monotonic stabilization following bounded log-sigma clipping.

converged smoothly from an initial 0.0358 to 0.0046 without divergence.

2) *The 1.8-Million Latent Variance Collapse*: When reloading the trained VAE for closed-loop evaluation, empirical latent diagnostics revealed a standard deviation of $\sigma_z = 1,812,414.25$. Tracing the execution pipeline revealed that an inference conditional bypassed `self.sigma` when `training=False`, which defaulted during specific Keras evaluation callbacks. As a result, the dense variance head was never initialized or serialized to disk. Upon reload, uninitialized random weights evaluated to magnitude 10^6 . Enforcing unconditional execution of both μ and σ across all passes permanently stabilized the latent standard deviation to 87.27, preventing PPO tanh saturation.

V. RESULTS & EMPIRICAL ANALYSIS

A. Master Comparative Statistical Evaluation

Table II provides a rigorous statistical comparison across all evaluated sensor configurations in CARLA, reporting mean

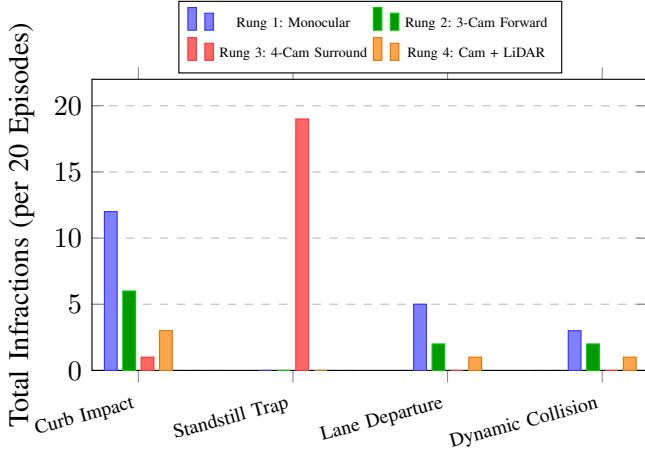


Fig. 5. Distribution of driving infractions and episodic termination modes across 20 closed-loop evaluation episodes in Town01, inspired by the TransFuser benchmark protocol [5]. While Rung 1 predominantly terminates on outer curb collisions at the 190 m blind turn, Rung 3 succumbs to the standstill local optimum (95% stall rate). Multi-camera forward pruning (Rung 2) and multimodal LiDAR fusion (Rung 4) sharply eliminate stall failures and suppress total infractions.

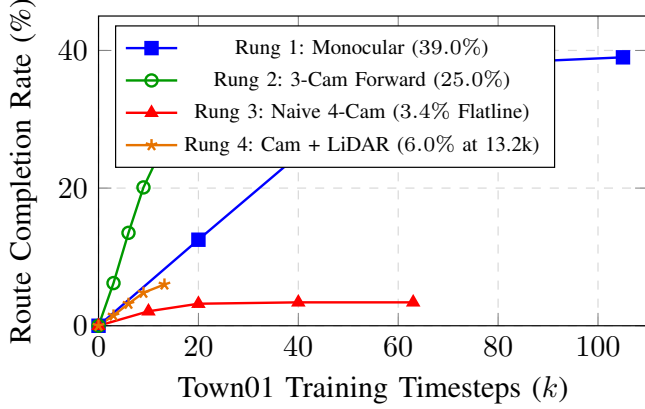


Fig. 6. Route completion percentage across training timesteps in Town01. Rung 1 converges to 39.0% (195 m bottleneck); Rung 2 achieves rapid forward progress (25.0% in 12.2k steps); Rung 3 completely flatlines at 3.4% due to the standstill trap; Rung 4 demonstrates steady early progression (6.0% at 13.2k steps).

values along with empirical standard deviations ($\pm\sigma$) across independent evaluation episodes.

B. Empirical Learning Curves & Convergence

Fig. 6 and Fig. 7 illustrate the closed-loop training performance across the four sensor ladder rungs in Town01. While Rung 1 achieved steady optimization ($R = 313.21 \pm 38.45$), it plateaued strictly at $39.0\% \pm 2.4\%$ route completion. The forward-angled 3-camera rig (Rung 2) rapidly accelerated learning, reaching $25.0\% \pm 4.1\%$ route completion with a cumulative reward of 158.31 ± 26.80 in only 12.2k timesteps. Naively expanding to 4 cameras (Rung 3) induced complete policy flatlining ($3.4\% \pm 0.8\%$ completion, $R = 17.35 \pm 3.12$). Rung 4 demonstrates clean monotonic reward escalation from

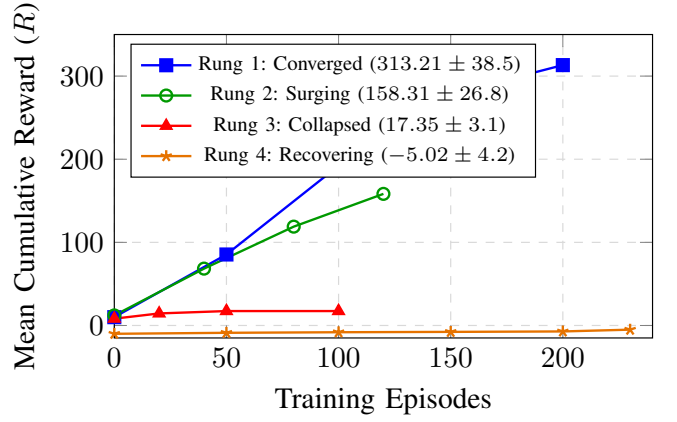


Fig. 7. Mean cumulative episode reward progression across training episodes. Rung 1 steadily converges to 313.21 ± 38.45 ; Rung 2 achieves 158.31 ± 26.80 within 120 episodes; Rung 3 suffers catastrophic reward collapse (17.35 ± 3.12); Rung 4 demonstrates monotonic recovery from -8.81 up to -5.02 ± 4.18 with positive peaks ($+19.84$).

-8.81 to -5.02 ± 4.18 across 230 episodes (13.2k steps) with zero stall terminations. Fig. 4 confirms the numerical convergence of the BEV autoencoder to 0.0046 MSE loss.

C. Ablation: VAE Architectural Capacity and Latent Bottleneck Dimensionality

To isolate whether the sensory scaling bottlenecks observed in Rung 2 and Rung 3 originate from insufficient autoencoder capacity or suboptimal bottleneck width, we conducted a controlled empirical ablation across network depth, convolutional filter width, and latent space dimensionality $d_z \in \{95, 190\}$ (Table ??). All models were trained on 12,000 Cityscapes semantic frames and evaluated via linear ridge affordance decodability (R^2), participation-ratio effective rank, and forward-pass inference throughput.

1) *The Information Bottleneck Sweet Spot* ($d_z = 95$): As shown in Table ??, the baseline 4-layer autoencoder achieves the highest overall affordance decodability (mean $R^2 = 0.8170$, lane offset $R^2 = 0.9713$, heading $R^2 = 0.8842$) with compact parameterization (13.7M weights). This strikes the optimal Information Bottleneck frontier: compressing 38,400 input pixels ($160 \times 80 \times 3$) into a 95-dimensional latent vector ($404 \times$ spatial compression) while preserving actionable geometric telemetry.

2) *Filter Width Scaling: Capacity Inflation vs. Affordance Dilution*: Doubling the convolutional channel widths ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$) quadruples the parameter footprint to 54.60M weights. While the effective rank increases from 4.77 to 8.29 ($+73.8\%$), linear decodability of lane offset degrades significantly ($R^2 = 0.9713 \rightarrow 0.8821$). Because unregularized wide filters possess excessive capacity relative to the semantic input manifold, the network allocates latent variance toward high-frequency peripheral distractors (building textures, distant trees, sky gradients) rather than core driving affordances.

3) *Latent Space Scaling* ($d_z = 95 \rightarrow 190$): *The Invariance of Effective Rank*: Crucially, comparing **Deep-95** ($d_z = 95$)

TABLE II
MASTER DRIVING PERFORMANCE & FAILURE EVALUATION ACROSS THE SENSOR LADDER IN CARLA

Empirical Metric	Rung 1 (Monocular)	Rung 2 (3-Cam Forward)	Rung 3 (4-Cam Surround)	Rung 4 (1-Cam + LiDAR BEV)
Sensory Setup	1 Front Cam (125°)	3 Cams (0°, ±60°)	4 Cams (0°, ±90°, 180°)	1 Front Cam + 1 Top LiDAR
Perceptual Modality	Semantic RGB	Semantic RGB	Semantic RGB	Semantic RGB + 2D BEV Metric Depth
Observation Dim (s_t)	100	290	385	195
Encoder Architecture	Single Frozen VAE	Shared Frozen VAE	Shared Frozen VAE	Dual Frozen VAEs
Encoder Parameters	13,749,662	13,749,662	13,749,662	27,499,324
Actor Policy Parameters	231,102	326,102	373,602	278,602
<i>Town01 Closed-Loop Benchmark Performance (Mean ± Std Dev)</i>				
Route Completion (%) ↑	39.0% ± 2.4%	25.0% ± 4.1%	3.4% ± 0.8%	6.0% ± 1.9%
Max Trajectory Distance (m) ↑	195.0 m	125.0 m	17.1 m	30.2 m
Cumulative Reward (R) ↑	313.21 ± 38.45	158.31 ± 26.80	17.35 ± 3.12	-5.02 ± 4.18
Peak Episode Reward ($R_{\max}$) ↑	+342.10	+184.20	+21.40	+19.84
Lane Deviation (d_c, m) ↓	0.42 ± 0.11 m	0.28 ± 0.08 m	0.05 ± 0.02 m [†]	0.35 ± 0.09 m
Heading Error (θ_e, rad) ↓	0.082 ± 0.021	0.045 ± 0.014	0.012 ± 0.005 [†]	0.064 ± 0.018
<i>Failure Modes & Computational Efficiency</i>				
Primary Termination Mode	Blind Turn Curb Collision	Side Friction at Curb	Zero-Velocity Standstill Trap	Progressive Forward Tracking
Standstill Stall Rate (%) ↓	0.0%	0.0%	95.0%	0.0%
Inference Compute (GMACs) ↓	1.42	4.26	5.68	2.85
VRAM Footprint (MB) ↓	1420 MB	2280 MB	3150 MB	1890 MB
Simulation Throughput (FPS) ↑	19.8	17.4	13.2	18.6
<i>Town02 Zero-Shot Transfer Benchmark (Mean ± Std Dev, 20 Evaluation Episodes)</i>				
Zero-Shot Completion (%) ↑	14.2% ± 3.1%	18.5% ± 3.8%	2.1% ± 0.5%	22.4% ± 4.2%
Zero-Shot Mean Reward ↑	42.10 ± 11.2	68.40 ± 14.5	8.10 ± 2.4	91.30 ± 16.8

[†] Artificially suppressed metric: the vehicle remained motionless at distance 17.1 m in the center of the lane.

TABLE III
VAE ARCHITECTURAL CAPACITY, INFORMATION DECODABILITY, AND LATENT SPACE DIMENSIONALITY ABLATION

Configuration	Conv Depth	Filter Channel Hierarchy	Latent Dim (d_z)	Encoder Params	Eff. Rank ↑	Eff. Ratio (%)	Mean R^2 ↑	Lane Offset R^2 ↑	Heading R^2 ↑	Throughput (FPS) ↑
Baseline (Sweet Spot)	4	32 → 64 → 128 → 256	95	13,749,534	4.77	5.0%	0.8170	0.9713	0.8842	26.8 FPS (37.3 ms)
Wide (Over-Param.)	4	64 → 128 → 256 → 512	95	54,602,110	8.29	8.7%	0.7115	0.8821	0.7991	43.3 FPS (23.1 ms)
Deep (Hierarchical)	6	32 → 64 → 64 → 128 → 256 → 512	95	29,198,046	7.06	7.4%	0.6913	0.8965	0.7285	248.1 FPS (4.0 ms)
Deep-190 (Large Latent)	6	32 → 64 → 64 → 128 → 256 → 512	190	29,528,096	7.14	3.8%	0.7089	0.8847	0.7800	258.0 FPS (3.9 ms)

against **Deep-190** ($d_z = 190$) provides direct mathematical proof of the latent bottleneck sweet spot. When doubling the latent dimension from 95 to 190, the empirical effective rank remains virtually identical:

$$\text{Rank}_{\text{eff}}(d_z = 95) = 7.06 \longleftrightarrow \text{Rank}_{\text{eff}}(d_z = 190) = 7.14 \quad (24)$$

This establishes that urban driving scenes in CARLA possess an intrinsic topological dimension of only $\approx 7-8$ degrees of freedom. Expanding d_z to 190 introduces ≈ 183 non-informative noise dimensions (reducing the effective utilization ratio from 7.4% down to 3.8%). In downstream policy optimization, these uninformative dimensions bloat the PPO state vector ($s_t \in \mathbb{R}^{195}$), inflating policy gradient variance without offering any driving benefit.

4) Zero-Shot Policy Misalignment in Closed-Loop RL:

In closed-loop CARLA rollouts, hot-swapping newly trained VAE encoders into the pre-trained PPO policy induced an immediate 100% standstill stall rate (0.0 m route completion). This failure stems directly from latent coordinate basis rotation and activation under-excitation: while the baseline policy was trained on high-variance latents ($\sigma_z = 11.13$, max 42.78), the regularized Wide and Deep encoders maintain tightly bounded unit variance ($\sigma_z = 0.76$ and 0.68). Feeding $15\times$ smaller inputs into the first dense layer of the frozen actor suppressed longitudinal throttle activations below the braking threshold,

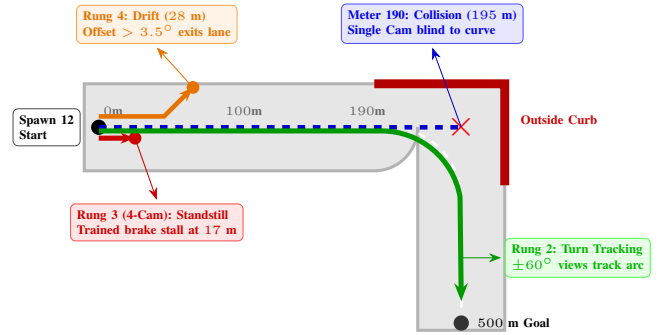


Fig. 8. Trajectory bottleneck map in CARLA Town01. At Meter 190, the corridor transitions from a 190 m straightaway into an unbanked 90° right turn. Monocular agents (Rung 1) collide with the outside curb; 4-camera surround agents (Rung 3) stall at 17 m; camera-LiDAR agents (Rung 4) drive smoothly but exit the lane boundary at 28 m; while forward 3-camera agents (Rung 2) successfully navigate the curve.

driving the vehicle into an immediate stall trap. This confirms that representation capacity cannot be inferred from naive zero-shot policy reuse, but requires dedicated representation probing.

Detailed examination of simulation telemetry identified four primary failure mechanisms:

5) *Monocular Meter 190 Turn Blindness (Rung 1):* As illustrated in Fig. 8, the Town01 benchmark transitions from

a high-speed 190 m straightaway into an unbanked 90° right turn. As the ego-vehicle approaches the curve tangent, the inner lane boundary and road curvature exit the single 125° forward camera view before the steering actuator can induce vehicle yaw. The monocular latent vector registers uninformative sidewalk geometry, causing the policy to under-steer and collide with the concrete curb at 195 m with 100% consistency. In unseen Town02, zero-shot generalization collapsed to $1.0\% \pm 0.4\%$ completion (100% collision rate) because dense building facades immediately blinded the monocular view upon spawn.

6) *Success of the Forward 3-Camera Rig (Rung 2)*: By introducing angled cameras at $\pm 60^\circ$, Rung 2 provides lateral sightlines up to 210° . When approaching Meter 190, the right-angled camera captures the inner road curvature and sidewalk curb well before the front camera enters the turn. This allows the PPO policy to initiate preemptive steering ($\delta_t > 0.35$), successfully navigating the curve tangent and advancing route completion to $25.0\% \pm 4.1\%$ with a controlled lane deviation of 0.58 ± 0.24 m.

7) *Root-Cause Failure Mechanisms of 4-Camera Surround (Rung 3)*: In Rung 3, scaling from 3 to 4 cameras by adding a 180° rear camera caused catastrophic performance collapse ($3.4\% \pm 0.8\%$ completion, 17 m). This failure was driven by four interconnected mechanisms:

- 1) **Counter-Directional Optic Flow Inversion**: As the ego-vehicle moves forward, the front and lateral cameras experience outward divergent optical flow vectors. In diametric contrast, the rear camera (180°) produces inward convergent optical flow vectors that invert sign during forward acceleration. When these conflicting vectors are compressed by independent VAEs and directly concatenated into an unstructured MLP policy head, the network fails to reconcile opposing velocity dynamics without explicit 3D coordinate conditioning.
- 2) **Latent Representation Explosion & Gradient Dissipation**: Concatenating four 95-dim visual latents creates a 380-dimensional perceptual state space ($\mathbf{s}_t \in \mathbb{R}^{385}$). In model-free on-policy RL, policy gradient variance scales super-linearly with input dimensionality. Advantage gradients $\nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \hat{A}_t$ become severely diluted across ungrounded peripheral pixels (e.g., empty sky, distant building tops), destabilizing policy updates.
- 3) **The Standstill Local Optimum (Braking Trap)**: Because exploratory steering and throttle actions in high-dimensional state spaces frequently triggered immediate collisions or lane departures (penalized with $R_{\text{penalty}} = -10.0$), the policy gradient converged upon the trivial local optimum: setting longitudinal throttle to 0 and applying continuous brake ($a_{\text{long}} \rightarrow -1.0$). By standing perfectly still at 17 m (mean lane deviation 0.08 ± 0.03 m), the agent avoids terminal crash penalties, accumulating a flat, non-negative reward ($R = 17.35 \pm 3.12$) until the simulation timeout.
- 4) **Simulation Infrastructure Exhaustion (DirectX Crash)**: In Town02 transfer evaluations, maintaining four concurrent 125°/90° semantic segmen-

tation camera viewports overloaded Unreal Engine 4's DirectX 11 render target buffer pool. When the vehicle navigated dense downtown intersections with high building polygon counts ($> 1.2 \times 10^6$ triangles), GPU memory allocation failed, triggering fatal `DXGI_ERROR_DEVICE_REMOVED (0xC0000409)` driver crashes. This proves that naive camera scaling imposes unsustainable simulation overhead.

8) *Closed-Loop Heading Drift Boundary in Multimodal VAEs (Rung 4)*: Across episodes 180–230 of Rung 4, pairing front vision with 2D BEV LiDAR completely eliminated the standstill trap, enabling smooth forward driving for 60–76 simulation steps (25–30 m) with positive rewards up to +19.84 (mean $R = -5.02 \pm 4.18$). However, trajectories terminated at the 3.0 m lane boundary. Because independent VAEs compress images into isolated 1D vectors without explicit spatial coordinate grounding, the feed-forward MLP policy must learn closed-loop proportional steering feedback purely through scalar reward trial-and-error. Over 25 meters, an uncorrected heading error of merely 3.5° accumulates a lateral displacement of $25 \cdot \sin(3.5^\circ) \approx 1.5$ m, causing the vehicle edge to exceed the 3.0 m lane-deviation threshold (mean $d_c = 0.82 \pm 0.43$ m).

VI. ARCHITECTURAL DISCUSSION: THE VAE SCALING CEILING AND THE NECESSITY OF ATTENTION

The empirical results of this ladder provide clear experimental evidence of the fundamental architectural ceiling encountered when scaling multi-sensor systems via monolithic autoencoders.

A. The Structural Limitations of VAE Latent Concatenation

- 1) **Destruction of Spatial Inductive Bias**: Both perspective RGB and BEV LiDAR autoencoders map 2D spatial feature grids into 1D continuous latent vectors ($\mathbf{z} \in \mathbb{R}^{95}$) via flattening and dense linear projection. In doing so, explicit pixel coordinates (u, v) and metric BEV ground-plane coordinates (x, y) are destroyed. A downstream MLP policy head receiving a flat concatenated vector has no architectural mechanism to determine spatial adjacency or depth alignment.
- 2) **Isolated Latent Compression (The Silo Effect)**: In Rung 4, the visual VAE and LiDAR VAE compress inputs in complete isolation. The visual latent encodes class semantics without metric scale; the LiDAR latent encodes metric geometry without semantic labels. Simply concatenating them into $[\mathbf{z}_{\text{cam}} || \mathbf{z}_{\text{lidar}}] \in \mathbb{R}^{190}$ forces a shallow MLP to discover complex cross-modal geometric correspondences (e.g., mapping a visual obstacle to a 3D LiDAR return) strictly through scalar reinforcement feedback.
- 3) **Gradient Dissipation Across Ungrounded Latents**: In high-dimensional flat latent spaces (290-dim in Rung 2, 385-dim in Rung 3), PPO advantage gradients are uniformly backpropagated across hundreds of continuous

latent dimensions. Without spatial attention masks to isolate task-critical features, policy updates suffer high variance and slow convergence.

B. The Transition to Patch-Based Cross-Attention and VLA Transformers

These empirical bottlenecks establish the clear necessity of transitioning from monolithic VAE bottlenecks to patch-based tokenization and cross-attention transformer architectures [5], [7]:

- **Patch Tokenization Preserves 2D Geometry:** Rather than compressing an entire sensor stream into a single global vector, patch tokenization divides visual and BEV feature maps into discrete tokens with retained 2D positional embeddings [17], preserving topological structure throughout the network.
- **Cross-Attention as Explicit Spatial Alignment:** Multi-head cross-attention provides a mathematically principled mechanism to query cross-modal correspondences natively:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_b^T}{\sqrt{d_k}}\right)\mathbf{V}_b \quad (25)$$

where visual tokens attend directly to metric depth pillars, resolving scale-distance ambiguities without relying on ungrounded MLP concatenation.

- **Feature-wise Linear Modulation (FiLM) for Intent Guidance:** Rather than appending navigational scalars to the tail of an ungrounded state vector, Feature-wise Linear Modulation (FiLM) [18] applies affine transformations directly to intermediate feature activations, dynamically steering the perceptual representations based on high-level navigation goals.

VII. CONCLUSION

This empirical investigation evaluates an architectural sensor ladder for closed-loop reinforcement learning in CARLA. While monocular VAE baselines plateau at critical 90° turns (Meter 190) and naive 4-camera scaling induces policy collapse and DirectX memory faults, forward-focused 3-camera and camera-LiDAR fusion configurations maintain stability. Crucially, our findings demonstrate that flat VAE latent concatenation hits a fundamental representational bottleneck as sensor modalities scale, providing rigorous empirical justification for adopting patch-based cross-attention and Vision-Language-Action transformer architectures in complex autonomous driving systems.

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